



Using Interaction and Quantitative Analysis to Examine the Effects of Video Modeling on Play of a Preschooler with Autism

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Accepted: 9 May 2024

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Abstract

Play, positioned powerfully in many cultures as a human right, is massively heterogeneous and resists a one-size-fits-all definition. Compared to typically developing individuals, researchers have characterized play in some individuals with autism as repetitive, stereotypical, less imaginative, and isolated. As such, the effectiveness, acceptability, and overall value of play-based interventions for individuals with autism have focused on discrete skills that facilitate normative play as opposed to the essence of play, such as playfulness and joy. The purpose of the current study was to use a mixed-method analysis to examine the effects of a video modeling-based play intervention implemented using a single-case multiple-probe design across playsets for a preschooler with autism. Visual analysis indicated that video modeling alone was not effective in enhancing scripted play, but combining video modeling with in situ modeling improved the participant's explicitly taught play. Our interaction analysis highlights five themes, providing nuanced insights into the participant's play that would not be otherwise captured through quantitative methods. Implications of adopting mixed-method analytical approaches to play in autism are also discussed.

Keywords Autism · Play · Interaction analysis · Mixed-method analysis · Video modeling

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Introduction

Among individuals with autism, play has been characterized as atypical, repetitive, stereotypical, less imaginative, and isolated compared to typically developing individuals (Jarrold, 2003; Jordan, 2003). Perhaps because of these unique forms of play expression, numerous studies have investigated teaching and “improving” play in children with autism through interventions (Freeman & Kasari, 2013; Lang et al., 2009). These studies recognize that play is crucial in the development of physical, cognitive, social-emotional, language, and self-regulation skills (Hughes, 2021; Siraj-Blatchford, 2009). Nonetheless, play remains heterogeneous and resists a one-size-fits-all definition (Souto-Manning, 2017). Broadly speaking, play can be characterized by activities that are (a) not functional, (b) spontaneous and voluntary/self-chosen, (c) exaggerated or distinct from typical activity, (d) guided by rules and patterns but done with imagination and without rigidity; (e) and initiated from a state of safety and spare energy (Burghardt, 2012; Gray, 2019).

In contrast to the inherent diversity within play, many play-based interventions for children with autism focus on teaching specific game rules or behaviors that would be perceived as ‘typical’ play or decreasing ‘atypical’ play (Dijkstra-de Neijis et al., 2021; Gibson et al., 2021; Kent et al., 2020). And within those interventions, the assessment and analyses of play in children with autism heavily rely on quantitative measures such as counting predetermined play actions (e.g., flipping over the cards in a card game; see Bernard-Opitz et al., 2004); verbalizations related to play (e.g., “It’s my turn.”; see Kamps et al., 2015); or normed social behavior (e.g., eye contact; see Chang et al., 2016). One could argue that these systematic and explicit instructions are necessary to support children with autism. Researchers and practitioners have continuously emphasized the importance of consistent, discrete, systematic instruction to promote maximum outcomes for individuals with autism (Knight et al., 2013; Smith, 2001). Current play interventions may be following the best practice of a discrete formula of systematic instruction (e.g., modeling and prompting) with specific learning objectives (e.g., saying “my turn”).

Video Modeling (VM) is one such method used in play interventions for children with autism (MacDonald et al., 2005; Maione & Mirenda, 2006). VM facilitates learning for children with autism by showing video clips or animated graphics of peers, adults, or themselves displaying or enacting the learning objective (McCoy & Hermansen, 2007). The utilization of VM in children with autism offers notable strengths primarily in terms of the degrees of individualization allowed in creating the video model based on individual needs (Ayres & Langone, 2005) and the time and cost efficiency achieved through development in consumer electronics and software (i.e., mobile phones) repeated use (Graetz et al., 2006). In addition, VM can provide abundant visual cues optimal enough to sustain the attention of some learners with autism (Charlop-Christy et al., 2000). VM has been acknowledged as an evidence-based practice for children with autism (Hume et al., 2021) and has proven effective for teaching a range of skills such as adaptive skills (see Hong et al., 2016), mathematics (see Wright et al., 2020), and social communication (see Qi et al., 2018). Consequently, many

studies have examined the effects of VM on play skills for children with autism such as decreasing repetitive play (e.g., Paterson & Arco, 2007) and increasing play-related verbal interactions and actions (e.g., D'Ateno et al., 2003; Dueñas et al., 2019; Nikopoulos & Keenan, 2003) through single-case experimental design (SCED).

SCED allows researchers to investigate interventions that are individualized and adapted for diverse samples acting as cases (Ledford et al., 2023) which can be especially useful for constructs such as play. However, relying solely on quantitative assessments may over-simplify play and fail to capture the unique experience of play in individuals with autism (Kasari et al., 2013). Adding a qualitative approach to data analysis may be one way to help address some of the limitations of measuring intervention effects solely based on quantitative data while also respecting the complexity and the human nature of play. In particular, qualitative work alongside SCEDs ensures space to consider alternative points of growth in play not covered by the SCED's pre-selected dependent variables, crucial to honoring a diversity of play-based contributions and setting the stage for more inclusive and responsive future research. Moreover, qualitative analysis has been a long-respected method to understand complex human behavior such as play (e.g., Chen, 2022; DeLiema et al., 2019). Prior qualitative work includes in-depth documentation of linguistic play in sibling culture (De León, 2007); children's negotiation of civic action during play (Adair et al., 2017); children's modification of the roles and rules of pretend play in a science classroom (DeLiema et al., 2021); and children's efforts to sustain and contest make-believe play (Sidnell, 2011).

For individuals with autism, qualitative inquiries that focus solely on their behavior and not as a comparison for typically developing behavior have great potential for highlighting and promoting neurodiversity, especially in areas such as play and social interaction. For example, in the past, "direct eye contact" has been utilized as a diagnostic indicator and taught as a social skill in autism research. However, emerging insights have revealed that eye contact without context fails to acknowledge and respect cultural and neuro-diversity (de Leeuw et al., 2020; Ma et al., 2021). Recent investigations in autism research and human behavior call for the need to comprehend individual behavior within contexts, culture, and identity especially social interactions using qualitative methodologies such as interviews (e.g., Perepa, 2014), content analysis (e.g., Trevisan et al., 2017), and interaction analysis (e.g., Dindar et al., 2017; Korkiakangas & Rae, 2014). Qualitative approaches offer an opportunity to understand social behaviors among individuals with autism and help identify the uniqueness of each individual, especially in personal contexts such as play.

Autism research using quantitative methodology has advanced the field such as offering us evidence-based practices that show strong effects for individuals with autism (e.g., Hume et al., 2021). However, focusing solely on quantitative methods has also led to many individuals with autism expressing discontent regarding the lack of consideration toward autistic autonomy and respect for neurodiversity (den Houting et al., 2021). Combining qualitative and quantitative methods may enhance our understanding of what is effective, acceptable, and valuable for individuals with autism within complex contexts such as play (Burney et al., 2023).

Therefore, in the current study, we use mixed-method analysis to investigate the impact of VM on increasing play of a preschool-aged child with autism. Specifically, we used visual analysis to assess the acquisition of predetermined play behaviors *quantitatively*, and interaction analysis of video to *qualitatively* investigate aspects of the child's play that may not be captured by the quantitative analysis.

Method

Participant

Daniel (pseudonym), a boy with autism, was recruited and selected based on the following inclusion criteria: (a) preschool-aged (i.e., 3–5 years old); (b) diagnosis of autism according to the Diagnostic and Statistical Manual of Mental Disorders (5th ed.; DSM–5; American Psychiatric Association [APA], 2013); and (c) need for social and play intervention per parent report. Daniel was 4 years and 5 months old at the beginning of the study and received a diagnosis of autism based on the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2; Lord et al., 2012) by a pediatric neurologist when he was 2 years old. Daniel lived at home with his biological mother and father, and a dizygotic twin sister, Abigail (pseudonym). Daniel's mother is Asian and his father is European American. Daniel's mother holds a master's degree and his father holds a bachelor's degree (other data related to socioeconomic status was not recorded). Daniel's twin sister, Abigail, is a typically developing girl and later participated as a play partner during generalization sessions of the study.

Daniel's information was collected by the first author through two 30-min unstructured parent interviews and one 30-min observation. Daniel's verbal expressive language included using one-word requests, some prelinguistic vocalizations such as babbling, and producing consonant sounds. Daniel also used an augmentative and alternative communication device (i.e., TD Snap) and gestures to communicate. Information from parent interviews and observation showed that play activities for Daniel involved repetitive manipulation of toys (e.g., press the button, push cars) and that he rarely spoke during play. Daniel's parents also reported that Daniel liked to watch videos, listen to music, and hum melodically. All participants were provided with informed consent before the start of the study—Daniel's parents provided written consent, and the first author obtained verbal assent at the beginning of every session from Daniel and also from Abigail during generalization sessions.

Setting and Materials

The study was conducted in a play area within Daniel's home, specifically in the living room of the basement. This area served as an exclusive space utilized by Daniel and his sister, Abigail, for daily learning and play activities. During the study, apart from Abigail's participation in the generalization phase, only Daniel and the first author were present in the play area without any interruptions. A Lenovo IdeaPad

Pro 13 laptop was used to play modeling videos. Two cameras (Canon Vixia HF R60) on two tripods were used to record all sessions of the study.

Playsets

Three playsets (i.e., Shark Dentist [Playset 1], Shark Bite [Playset 2], and Shaky Shark [Playset 3]) were used for this study (see supplementary material). The playset was informed by Daniel's interest in shark-themed toys, as gathered from parent interviews. All three playsets employed in the study can be characterized as interactive cause-and-effect toys. In Playset 1 ($5.9 \times 4 \times 3.7$ inches), the shark randomly bites (i.e., closes its mouth) when the player engages in the act of pushing on its teeth. In Playset 2 ($7.5 \times 5 \times 10.5$ inches), the shark randomly initiates biting when fish are pulled out from its mouth. Playset 3 ($8.5 \times 7.5 \times 11$ inches) triggers the shark's random biting when the treasures within its mouth are taken out for a certain duration. Among the three playsets, only Playset 3 produces a rattling sound, signaling the onset of the random biting action.

Modeling Videos

The first author created three 30-s modeling videos depicting two adult models playing with each playset. A shark-themed song (i.e., Baby Shark) that Daniel reportedly liked was embedded in the videos as intro music to possibly increase Daniel's interest. In the three videos, the first author and another adult played with a playset adhering to scripts. Each script included one-step play actions (e.g., push, pull, grab) with one-to-three-word verbalizations (e.g., "push," "your turn," "he bit me"). Across the three playsets, there were 12 scripted play actions and verbalizations (see Table 1). Daniel's parents agreed on the scripted play actions and verbalizations as learning goals for Daniel.

Experimental Design

A single-case multiple-probe design across playsets was used to assess the effects of video modeling on acquisitions of scripted play actions and verbalizations. We planned to first implement only the video model (*VM-Only*) and then add in situ adult live modeling if Daniel needed prompting. After observing stable low-level responses in three sessions for Playset 1 with *VM-Only*, we added in situ modeling (*VM + ISM*). Daniel demonstrated low-level responses across all three playsets for *VM-Only*, and *VM + ISM* was implemented for all playsets. Six data sessions were collected for each playset for *VM + ISM* due to the time constraints of the participating family. Generalization probes occurred 0–2 days after the completion of each playset's intervention. Maintenance probes were conducted 1 and 2 weeks after each playset's intervention.

Table 1 Play scripts in modeling videos

Playset	Verbalization	Action
Playset 1 Shark Dentist	Let's play	Move the shark (swimming movement)
	Push!	Push one tooth of the shark
	(1) Ohhh I'm safe!	Clap hands to celebrate
	(2) Ohhh he bit me!	Shake the hand that was bit
	Your turn!	Move the toy close to the other player
Playset 2 Shark Bite	My turn!	Move the toy close to him/herself
	Let's play	Open and close the mouth of the shark
	Get the fish	Pull out one object from the shark's mouth
	(1) Thank you	Wave the fish
	(2) Ouch!	Rub the hand with the other hand
Playset 3 Shaky Shark	Your turn!	Move the toy close to the other player
	My turn!	Move the toy close to him/herself
	Let's play	Make the shark wamble several times
	Save one!	Get one object from the shark's mouth
	(1) Yay! I got one!	Wave hands in the air
	(2) Ohhh time's up!	Shrug and spread one's hands
	Your turn!	Move the toy close to the other player
	My turn!	Move the toy close to him/herself

Procedure

The intervention procedure included four conditions: (a) baseline; (b) intervention; (c) generalization; and (d) maintenance. The details of the procedure are provided in Fig. 1.

Baseline

During baseline sessions, Daniel was sitting on the same side of the table as the first author and was presented with the playset and verbally instructed, "Let's play." The duration of each play session was 3 min. No experimenter-implemented reinforcement, prompting, or correction procedures were used.

Intervention

The intervention phase was implemented in the same setting as the baseline phase. During the *VM-Only* phase, Daniel was directed to watch the video. After watching the video twice, Daniel was given access to the playset depicted in the video. The duration of each play session was 3 min. For *VM-Only*, no experimenter-implemented reinforcement, prompting, or correction procedures were used. During the *VM+ISM* phase, Daniel was directed by the first author to watch the video from start to finish two consecutive times. Then the first author initiated the scripts following the same sequence as depicted in the video until the point where Daniel had

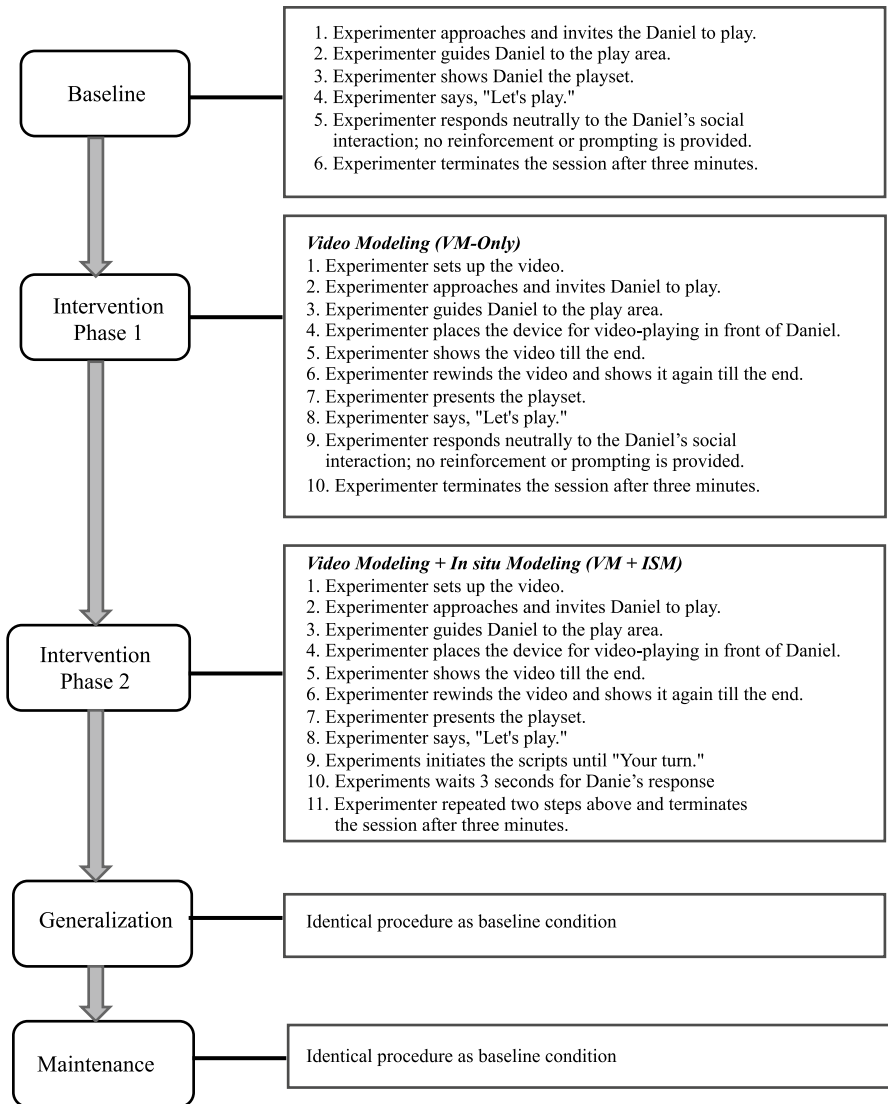


Fig. 1 Procedure summary

to respond. The first author waited three seconds for Daniel's response. For example, in Playset 1, after playing the video model two times, the first author presented the playset and said, "Let's Play!". Following this, the first author pushed one of the shark's teeth, simultaneously verbalizing the action with, "Push". Depending on the shark's reaction (i.e., the shark randomly closes its mouth), the first author then responded with either, "Oh, I'm safe!" or "Oh, he bit me!" and demonstrated corresponding play actions. After verbalizing, the first author paused, allowing a three-second window for Daniel's response. If Daniel did not respond, in situ live

modeling was repeated until the end of the 3-min play session. No experimenter-implemented reinforcement or correction procedures were used.

Generalization and Maintenance

During the generalization sessions, Daniel's sister, Abigail viewed the video with Daniel jointly. The procedure for generalization was the same as the baseline except Abigail replaced the role of the first author. The maintenance sessions were identical to baseline sessions, with the first author serving as the play partner.

Measurement and Data Analysis

Quantitative Data

Target behaviors for Daniel included scripted play actions and play verbalizations. *Scripted play action* was defined as any play action from the modeling video approximated by the child. A response was scored as correct if the play action was non-identical but served the same function. For example, the modeling video of Playset 2 depicted retrieving one fish at a time but retrieving multiple fish was still counted because it aligned with the objective of the game (i.e., retrieving fish). An incorrect response was defined as any play action that was not demonstrated in the modeling video or involved no interaction with the playset. For example, if the participant patted the head of the shark, an action not featured in the modeling video, or did not touch the playset during the play session, such responses were scored as incorrect. *Scripted play verbalization* was defined as any verbalization that approximated the video model verbalizations in context. For example, saying "fish" (instead of the full "get the fish" phrase) was still scored as correct because the objective (i.e., labeling fish) was retained. An incorrect response was defined as any play verbalization that was not demonstrated in the modeling video. It included instances where participants emitted verbalizations that were not modeled in the video (e.g., calling out a family member's name, signing) or where participants did not verbalize during the play session. Although there were certain steps toward the goal of each playset, because of the randomization of the toys used in each play set and to foster natural play as opposed to prescriptive play actions, no total score was determined. The frequency of the target behaviors was tabulated and no partial instances (e.g., half scores) were considered.

The first author tabulated the frequency of target behaviors (i.e., scripted play actions and verbalizations) exhibited by Daniel in each 3-min play session by watching recorded videos of the sessions. Consecutive repeated displays of the same target behavior by Daniel were counted as one because they occurred out of sequence and were not interactive. For instance, in Playset 1, if Daniel pushed the shark's teeth several times in succession without displaying another target behavior, the recorded frequency for that behavior was counted as one as opposed to the number of times Daniel pressed the shark's teeth. We used visual analysis (see Lane & Gast, 2014) to evaluate change in target play behaviors.

Qualitative Data

The qualitative data analysis involved a re-examination of the video recordings with the purpose of understanding how Daniel participated in play in ways not captured by our earlier quantitative analysis using interaction analysis (Derry et al., 2010; Jordan & Henderson, 1995). Interaction analysis involves observing and analyzing sequences, patterns, and qualities of interactions between individuals to unveil underlying processes, patterns, and meanings within a specific context (Levin & Brown, 2015). In this study, the specific context pertained to the play interactions between Daniel and the first author. We (i.e., the first, second, and third author) closely examined how Daniel and the first author calibrated their actions to one another and analyzed how Daniel and the first author's talk, gesture, gaze, and action contributed to subsequent multimodal interactions.

We created multimodal transcripts that represented and synchronized these multiple modes of communication, and then iteratively refined the transcripts during interaction data analysis sessions to ensure the transcripts represented the interaction with fidelity. Multimodal transcripts are documents that capture communication in multiple modes, which not only include traditional text-based transcripts but also incorporate additional elements such as pauses and intonation in verbal interaction, gestures (e.g., hand movements), facial expressions (e.g., smiles), and other non-verbal behaviors (Bezemer & Mavers, 2011). Examples of our multimodal transcripts are presented in Figs. 4, 5, 6, 7 and 8.

In our analysis, we focused on *how* Daniel and his play partner (i.e., the first author) managed the play activity without speculating on Daniel's intentions, private thoughts, or personal emotions. This involved a decision to identify video clips at the point in the study where Daniel had the most experience with the game and his play partner that would provide maximum insight into Daniel's approach to play. We selected the focal case for interaction analysis given that it is typical of the way Daniel played toward the end of the study and across all three playsets; this selection process, where the researcher focuses on a pattern or set of patterns common across a range of setting, is a common approach in conversation and interaction analysis (Hoey & Kendrick, 2017; Jordan & Henderson, 1995). Specifically, we selected the first maintenance session of Playset 2 (i.e., the session conducted 1 week after the conclusion of the intervention of Playset 2) as the focus of qualitative analysis. The selection of a maintenance session aimed to capture Daniel's maximum skill level and play experience compared to intervention and generalization sessions. After the themes were identified, we returned to the other maintenance sessions to confirm the existence of the five themes in the other videos. It is important to note that the five themes were also observed in Daniel's play across the other playsets' maintenance sessions.

Building on methodological tenets of interaction analysis (Hall & Stevens, 2015) and thematic analysis (Braun & Clarke, 2006, 2021), the qualitative data analysis process involved five stages employing an inductive process:

1. We (i.e., the first, second, and third author) each individually viewed the video of Playset 2's maintenance session, separately transcribed key moments, and described Daniel's approach to play.
2. We held collaborative data sessions in which we watched the video, paused and re-watched focal moments while sharing observations and engaging in negotiated agreement processes (Campbell et al., 2013) about how the participants managed the play activity.
3. We refined our multimodal transcripts and continued using them to articulate themes representing common interactional patterns in play that were not already captured by our SCED's dependent variables.
4. We finalized themes through a process of constant-comparative analysis (Glaser, 1965), chose which transcripts to use for dissemination as representative examples of each theme, and wrote the qualitative interaction analysis.
5. We viewed maintenance session videos of the other playsets, collectively discussed observations to ascertain whether the finalized themes were similarly evident.

In this way, we engaged with the value of interaction analysis to unpack five recurring ways that Daniel contributed to the play activity in that session. Importantly, this analysis was not intended to capture the frequency of these play processes. Indeed, holding back from engaging in frequency counts at the stage of work is critical to respecting the complex process of orienting to new phenomena in interaction and building a sense of which discursive settings might occasion those phenomena (de Ruiter & Albert, 2017; DeLiema et al., 2021; Schegloff, 1993).

Interobserver Agreement

A graduate research assistant (GRA) who majored in special education was trained to provide interobserver agreement (IOA) data. For IOA training, the first author and the GRA reached 95% (range 84–100%) agreement on three practice sessions, one session for each playset. Then the GRA independently coded data for randomly selected 33% ($n = 16$) of all 48 sessions across baseline, intervention, generalization, and maintenance (excluding sessions used for training). We calculated the percentage of agreement for IOA. The IOA between the first author and the GRA was 93% (range 77–100%) for scripted actions and 91% (range 76–100%) for scripted verbalizations.

Social Validity

Daniel's mother and sister, Abigail, shared their perspective on social validity after the intervention. After the generalization sessions, the first author presented Abigail with three emotion cards (i.e., a sad face, a neutral face, and a smiley face) to inquire about Abigail's experience. The first author also asked two open-ended questions about Abigail's favorite video and playset. Daniel and Abigail's mother watched the recorded videos of baseline and intervention sessions and assessed the

goal, procedure, and effectiveness of the study using a 5-point Likert-type scale ranging from strongly disagree to strongly agree. Daniel and Abigail's mother also answered an open-ended question to provide additional feedback (see supplementary material).

Results

Quantitative Results

Figure 2 depicts a line graph of Daniel's scripted play actions during baseline, intervention, generalization, and maintenance conditions across three playsets. Daniel demonstrated some scripted play actions during baseline (Mean=1, Range 0–2), but the actions remained at a stable low-level across playsets. During the *VM-Only* phase, Daniel's scripted play actions also remained low-level (Mean=0.78, Range 0–1) across playsets. Following the introduction of *VM+ISM*, the overall frequency of scripted play actions increased. The mean numbers of play actions were 10 (Range 6–19) for Playset 1, 11.17 (Range 9–13) for Playset 2, and 4.33 (Range 1–8) for Playset 3 with clear level changes. Daniel also demonstrated generalization of scripted play actions with his sibling Abigail, although with a lower level of occurrence compared to intervention (Mean=9 for Playset 1, Mean=9 for Playset 2, Mean=2 for Playset 3). No notable generalization of play actions was observed for Playset 3. Daniel's performances in the maintenance sessions were similar to the *VM+ISM* phase (Mean=17.5 for Playset 1, Mean=11 for Playset 2, Mean=11.5 for Playset 3).

Data for Daniel's scripted play verbalizations are displayed in Fig. 3. Mean number of scripted play verbalizations was 0.15 (Range 0–1) across playsets per session during baseline. During the *VM-Only* phase, frequency of scripted play verbalizations did not increase (Mean=0.67, Range 0–1). With the introduction of *VM+ISM*, Daniel's scripted play verbalization increased for Playset 1 (Mean=5.67, Range 2–16) and Playset 2 (Mean=11.17, Range 6–19), but there was no change for Playset 3 (Mean=0.33, Range 0–1). During generalization, the mean number of scripted verbalizations was 2.5 (Range 2–3) for Playset 1, 6 (Range 4–8) for Playset 2, but the Mean was zero for Playset 3. Daniel's performance for Playset 1 and Playset 2 was sustained across maintenance sessions (Mean=11 for Playset 1; Mean=11 for Playset 2). Daniel increased play verbalizations in the maintenance sessions for Playset 3 (Mean=3.5, Range 0–7).

Qualitative Results

Our interaction analysis led us to identify five themes that illustrate Daniel's learning and engagement in social interaction during play which were not captured by our predetermined dependent variables. The themes are (a) individual expression, (b) turn-taking, (c) repatterning, (d) joint immersion, and (e) set up / transition (see Table 2 for detailed description). These five themes were inductively identified from

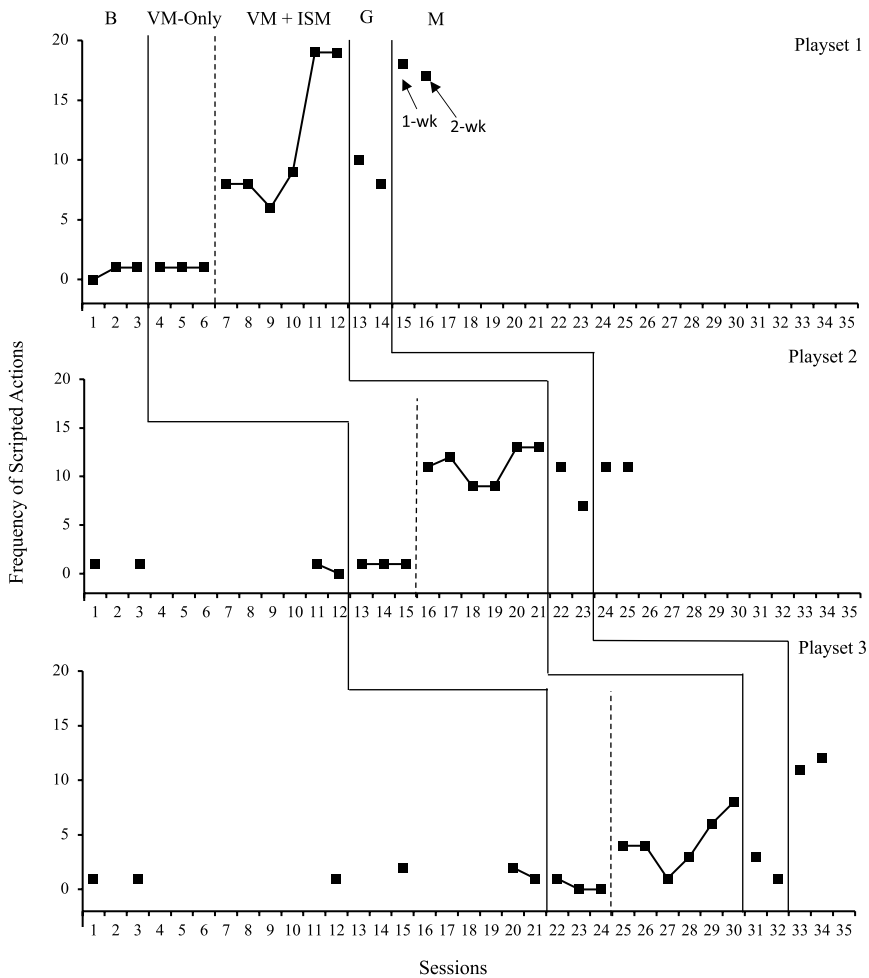


Fig. 2 Frequency of scripted actions across three playsets. *B* baseline, *VM* video modeling, *ISM* in situ modeling, *G* generalization, *M* maintenance

Playset 2's maintenance sessions. The five themes were also later observed in Daniel's play across all three playsets in maintenance sessions. We provide transcripts from a focal clip in Figs. 4, 5, 6, 7 and 8 (F4–F8) as examples. The transcripts display images in a sketch-artist style for anonymity with verbalization and action bubbles capturing the details of the moments.

Individual Expression

Daniel consistently enacted individual forms of expression throughout the shark game. In this instance, Daniel held onto the yellow fish after extracting it, tapped it on the table (F4.L03), pulled it close to his face for inspection (F4.

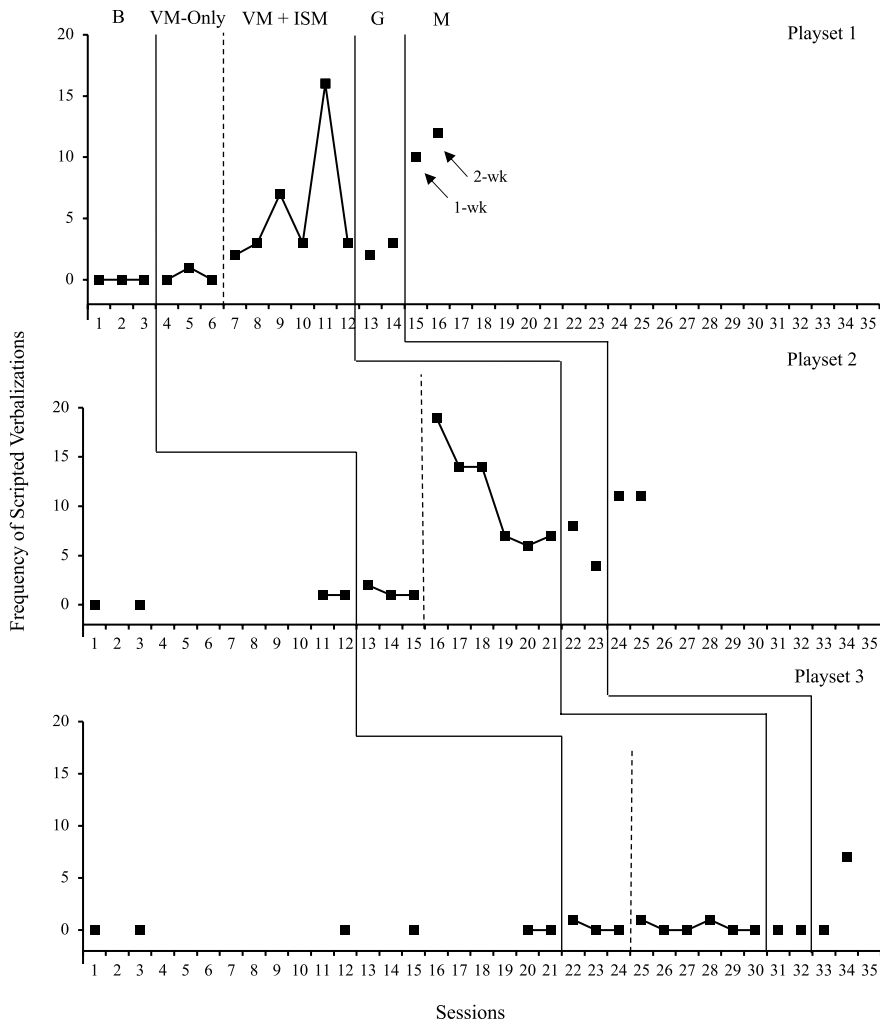


Fig. 3 Frequency of scripted verbalizations across three playsets. *B* baseline, *VM* video modeling, *ISM* in situ modeling, *G* generalization, *M* maintenance

L03–04), and then clutched it with both hands (F4.L08), even while attending to Zoe (i.e., the first author) as the turn shifted to her (F4.L07). In other instances of individual expression, Daniel sorted the fish on the table into their distinct shapes, moved the fish in unique swimming patterns, and strategically extracted particular fish from the shark’s mouth. These moves, which fell outside of the normative modeling in the play video, demonstrated Daniel’s original take on the game and how he found meaning within it.

Table 2 Themes and descriptions

Theme	Description
Individual expression	Daniel's unique ways of voluntarily expressing his own play behaviors during play activities. These are imaginative and playful ways Daniel approaches the game that move beyond and creatively extend the game's pre-defined rules and the game's actions directly presented in the modeling videos. These individual forms of expression are widespread across play contexts, for example, as children add unique vocalizations, ways of moving play objects, sorting play objects, etc. during a game
Turn-taking	Daniel's approach to taking turns during the game. In general, turn-taking is the process of selecting and enacting the order of different game players' distinct game actions. In contrast to improvised turn-taking in everyday conversation (Sacks et al., 1978) and improvised turn-taking in open-ended play (e.g., Garvey, 1974; Hoey, 2018), these turn-taking patterns are often prescribed by the rules of the game (e.g., when to hand off a turn or when to skip a turn). For Daniel, these turn-taking play activities might include anticipating when his turn is coming by making sense of other players' actions and/or intentions, taking action to lead the exchange of turns, relinquishing his own turn whether prompted or not, and following prompts from others to take his turn
Repatterning	Daniel's attempts to modify, change, or replace established play patterns in a way that is distinct from the patterns set up in the modeling videos. What makes this different from individual expression is that this repatterning has the potential to change/affect other players' game actions, strategies, goals, etc. Because these might have a bearing on the group's approach to the game, repatterning efforts invite others to negotiate the new approach, whether welcoming, ignoring, or resisting the change
Joint immersion	Daniel and his play partner improvisationally navigate play in such a way that both participants can attend to the same game event and/or outwardly communicate their own experience of that game event. These aspects of joint immersion were not required, elicited, or explicitly taught with the video models. In the moments of joint immersion, players may communicate similar experiences of that game event (e.g., both players laughing about a shark bite), or different experiences of the same game event (e.g., one player celebrating while another player laments a player winning the game). However, because both players are attending to and communicating about the same game event, these emergent actions are likely integral to the joint involvement, synchronous reactions, and enjoyment that comes from play. This category is distinct from parallel play, in which an individual is in close proximity to someone else who is engaging in the same task but without direct interaction between each other (Luckey & Fabes, 2005)
Set up/transition	Daniel's participation or behavior outside of or between play sessions that is not exactly necessary or required to participate in the core play (i.e., game) but when displayed facilitates transitions that increase the time spent and prepare for the play itself. This includes behaviors related to getting ready and helping to restart the play

Turn-Taking

When it came to turn-taking, Daniel often demonstrated precise timing, in this case pausing an individual form of expression (examining a yellow fish) immediately after Zoe's vocal statement about the upcoming turn change (F5.L04–05),

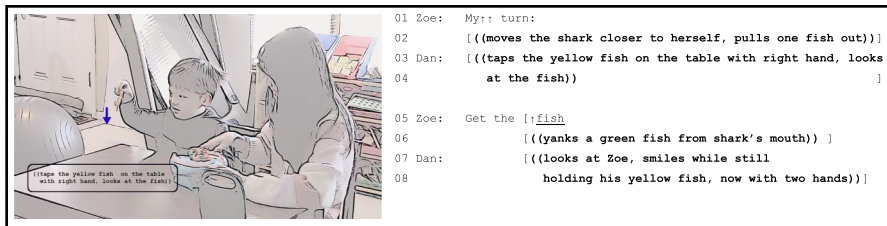


Fig. 4 Transcripts of example of individual expression

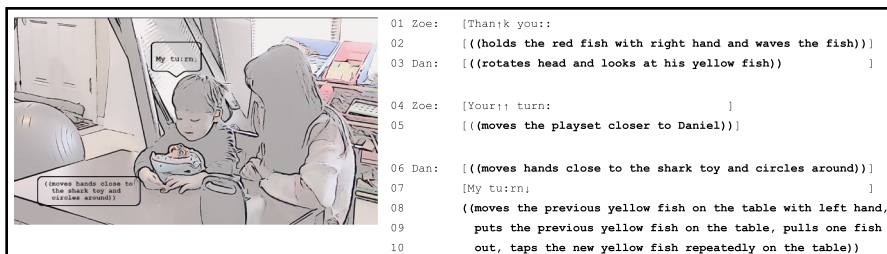


Fig. 5 Transcripts of example of turn-taking

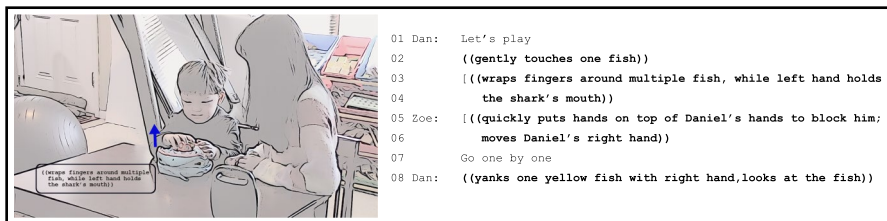


Fig. 6 Transcripts of example of repatterning

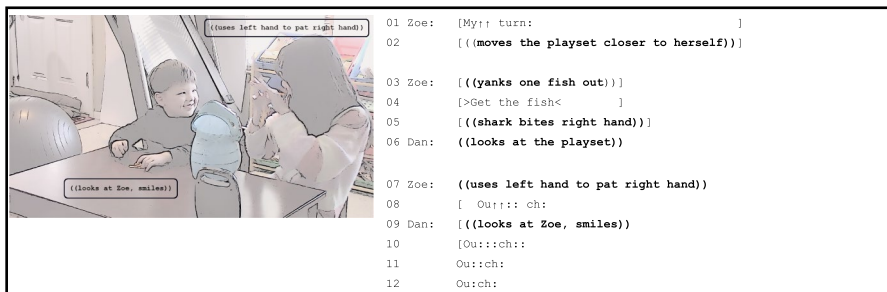


Fig. 7 Transcripts of example of joint immersion

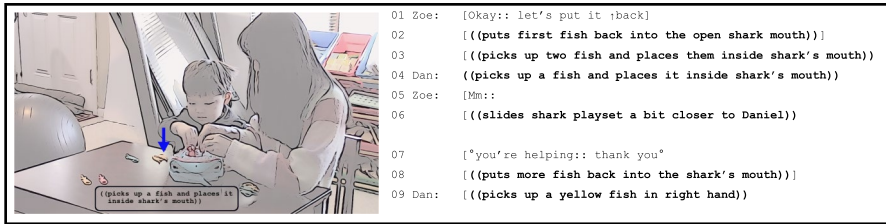


Fig. 8 Transcripts of example of set up/transition

participating in moving the shark toward himself (F5.L06), and then completing the previous turn (by placing his prior yellow fish on the table) before grabbing a new fish (F5.L08–10). These precise turn-taking actions were evident elsewhere, whether in Daniel anticipating his turn (e.g., reaching for the shark just as Zoe was finishing) or ending his turn by removing his grip from the shark, all of which signaled his precise attunement to the turn-taking dynamics of the game.

Repaterning

Daniel also contributed to the game in ways that repatterned the play activity for him and Zoe. In this instance, Daniel made a bid to alter the number of fish extracted from the shark. He started by touching one fish (F6.L02), extended his reach to the tops of multiple fish (F6.L03–04), received feedback from Zoe that physically blocked (F6.L05–06) and vocally course corrected his effort (F6.L07), and then returned to grab one fish (F6.L08). Daniel's bid in this instance was met with resistance from Zoe, and he adjusted his plan to follow the typical pattern in the game, but in other instances later in this play session, Daniel revised the pattern by grabbing multiple fish in a single turn and received no pushback from Zoe. With respect to this contribution, Daniel demonstrated both awareness of a tendency in play to follow agreed-upon patterns, but also enacted a vital and widespread feature of play, which is to re-pattern the activity to explore new possibilities.

Joint Immersion

Daniel further contributed to the play activity in ways that involved synchronizing his efforts with Zoe. In this instance of joint immersion, Zoe removed a fish from the shark's mouth (F7.L03–04), which triggered the shark's bite (F7.L05). Daniel immediately turned his gaze to the playset (F7.L06). Zoe began patting her bitten right hand (F7.L07), and she and Daniel looked directly at each other and both simultaneously vocalized "Ou:::ch::," all while Daniel smiled (F7.L08–10). This moment served as one of the many instances of Daniel and Zoe sharing their emotional experience of a salient game event as they were jointly immersed in the play activity.

Set Up/Transition

Finally, Daniel further contributed to the play activity in set up and transition moments. Here, after one round of the game had ended, Daniel contributed by picking up multiple fish (F8.L04 and L09) and returning them to the open mouth of the shark (F8.L04). Zoe noticed and acknowledged his contribution (F8.L05), moved the playset closer to Daniel so that he could contribute (F8.L06), and narrated her gratitude that Daniel was helping (F8.L07). As with the themes described above, these actions demonstrate Daniel's contributions to the game activity that were above and beyond what was modeled in the video and what was included as dependent variables in the SCED, providing an account of additional ways Daniel was contributing to play that would have been missed.

Social Validity

Results of the social validity assessment indicated a high level of acceptance and satisfaction with the study from Daniel's mother and sister. Abigail answered the social validity questions and mentioned that she enjoyed watching videos and playing with Daniel. She also stated her preference for Playset 3 (i.e., Shaky Shark). The results of the social validity questionnaire indicated that Daniel's mother agreed with the importance of intervention goals and the acceptability of procedures, and was satisfied with the outcomes. In the comment section, Daniel's mother expressed her desire for a longer study duration and suggested adding subtitles to modeling videos to potentially enhance targeted play behaviors (see supplementary material for details).

Discussion

The purpose of this study was to explore the impact of VM on teaching play behaviors to a preschool-aged child with autism through mixed-method analysis. The quantitative data indicated a combination of video modeling and in situ modeling showed potential to increase scripted play actions and verbalizations. The qualitative results, shaped by interaction analysis, highlighted the participant's spontaneous and emergent play behaviors not captured with the predetermined dependent variables. Implications from this study can inform future play-based interventions for individuals with autism, particularly in assessing play outcomes through mixed-method analysis and the importance of developing individualized play-based interventions.

Assessing Play Outcomes with Mixed-Method Analysis

In recent years, an increasing number of studies have shifted their focus toward unscripted play behaviors in play-based interventions for individuals with autism (e.g., Dueñas et al., 2019; Pane et al., 2022), as well as emphasizing the quality of

play within interventions. However, the measurement of play outcomes for individuals with autism within interventions remains predominantly quantitative (Dijkstra-de Neijs et al., 2021; Kent et al., 2020). For example, turn-taking within play scenarios has often been included as an intervention objective to facilitate social interaction development in individuals with autism (Rieth et al., 2014). Outcomes measured in these interventions are typically quantitative such as the number of turns taken within a specific time frame (e.g., Kim & Clarke, 2015; Therrien & Light, 2018). But the intricacies of human play require a deeper and more comprehensive understanding of turn-taking within play (Wiemann & Knapp, 2017). For example, in natural play, children and adults alike could purposefully “sneak in” an extra turn, often resulting in playfulness outside the game rules. Simplifying turn-taking to quantitative data without appreciating the contextual nuances in play may over-simplify the nature of play (e.g., enjoyment; Bottema-Beutel et al., 2022; Sicart, 2014).

This study employed a mixed-method analysis approach, combining both quantitative and qualitative assessments of play. First, we assessed the functional relationship between the intervention and scripted play behaviors through quantitative tabulation of pre-defined dependent variables. The quantitative analysis results provided support for the effectiveness of the current intervention in enhancing scripted play behaviors. The data plotted in the time-series line graph offered a depiction of the learning growth of Daniel such as the overall limited learning growth in scripted play verbalizations compared to the acquisition of play actions. Additionally, a decreasing trend was observed in the scripted play action of Playset 2. This decreasing trend could potentially be attributed to Daniel’s limited expressive language skills in comparison to the required verbalizations depicted in the video model.

Next, our qualitative interaction analysis allowed us to analyze participant- and context-specific aspects of play. For example, we quantified turn-taking through verbal cues (e.g., “My turn”) and actions (e.g., moving the playset closer to another player). Additionally, the *turn-taking* theme, identified through qualitative analysis, highlighted contextual behaviors vital for understanding how Daniel was learning to take turns within the games. These include Daniel anticipating his upcoming turn by interpreting the other player’s actions and/or intentions (e.g., rotating his head to look at the other player), initiating the exchange of turns (e.g., reaching out to the playset), and respecting the other’s turn (e.g., not reaching out to the playset until the other player finishes their turn). We argue that these behaviors may be stronger demonstrations of understanding turn-taking within play than the number of turns.

A qualitative analysis can add insight into the potential effects of an intervention and the participant’s emergent play behaviors not included in a quantitative analysis. Quantitative analysis of play-based interventions risks an over-emphasis on quantities of prescribed play behaviors. In autism research, there is a scarcity of work probing the various facets of play in individuals with autism, including rule negotiations, repatterning rules within social consent, failures and reactions to failures, as well as the experiences of boredom. Mixed-method analysis can help identify and explore socially situated and contextualized learning processes and holds promise for advancing our future research endeavors (McCrudden et al., 2019). It offers a way to evaluate interventions based on the true nature of play and is inclusive of neurodivergent ways of play. We should also remember that Daniel’s play will differ

from that of other individuals. The inclusion of qualitative analysis will allow the field to understand and share what it means to play and enjoy play for individuals.

Developing Individualized Play-Based Interventions

Play in individuals with autism is often depicted as delayed or deviating from the play development trajectories in typically developing individuals. For instance, many studies compare individuals with autism to those without autism and evaluate their play (e.g., Charman et al., 1997; Rutherford et al., 2007). Other times, play for individuals with autism is used only as a medium to facilitate social interaction with typically developing peers as opposed to an activity that brings joy (e.g., Henning et al., 2016; O’Keeffe & McNally, 2023). Play-based interventions based on a deficit or deviated approach may overlook the heterogeneous nature and individual expression of play which is the essence of play (Howard & McInnes, 2013).

Our decision to investigate the intervention’s effectiveness based on scripted play, as well as those behaviors that were not predetermined, was attributed in part to the emphasis on an individualized approach in play-based intervention development. While a substantial body of literature supports the use of VM to enhance the social behavior of individuals with autism (Bellini et al., 2007; Lee, 2015; McCoy et al., 2016), the decision to use VM was primarily influenced by Daniel’s interest in watching videos. Anecdotal experience suggests that the shark-themed playsets and the shark-related song in the modeling videos notably enhanced Daniel’s joy and engagement during intervention. For example, playfulness demonstrated by Daniel throughout the intervention may be one explanation for Daniel’s engagement even with the absence of a reinforcement procedure (e.g., social praise, tangible preferred item).

In our study, the initial approach (i.e., *VM-Only*) was chosen based on our observation of Daniel’s preference for learning and entertaining through videos, as indicated in the pre-intervention parent interview. However, VM alone was ineffective in enhancing scripted play behaviors initially and the adjusted approach—a combination of VM and ISM—proved to be more effective. While some studies suggest that VM can be more time-efficient and yield better generalization compared to ISM (Charlop-Christy et al., 2000), our research adds to the current literature that a combination of different modeling strategies can serve as a more effective alternative when a particular modeling approach may not be sufficient support for some individuals with autism. In a recent review, Bak and colleagues (2023) found that participants included in intervention studies were from a very homogenous population and individuals with communication difficulties were seldom included in intervention research outside of investigations for augmentative and alternative communication. Researchers should keep in mind that although evidence-based interventions may have a strong empirical base for effectiveness, some individuals with autism may need additional support. Additional support such as ISM can enhance learning opportunities through in-person engagement and provide researchers with the flexibility of including additional prompting for some individuals with autism.

Playfulness in Play Interventions

Existing play-based interventions often miss capturing the playful manner in which individuals with autism approach play. In studies aimed at teaching children with autism to play, individuals with autism may inadvertently be learning scripted and sequenced behaviors as opposed to enjoying play (O'Brien et al., 2000). The effectiveness of maintaining and generalizing these acquired behaviors into play, then, becomes questionable (Kasari et al., 2013). Within our *repatterning* theme, Daniel changed the rules illustrated in the modeling videos, enhancing the playfulness of the experience (e.g., pushing multiple shark teeth in Playset 1; getting multiple fish in Playset 2). Such adaptability is one of the key characteristics for recognizing the sheer diversity of what constitutes playfulness (Ruberg, 2017). In another theme, *joint immersion*, Daniel shared both congruent (e.g., laughing together) and contrasting play experiences (e.g., expressing joy when another player fails). Such examples further emphasize the importance of a playful nature in play interventions for children with autism. The delivery of play-based interventions for individuals with autism should not only concern outward play behaviors, but also analyze, to an extent, joy and interactive play experiences, including playfulness experienced by the child with autism (Gordon, 2014; Skaines et al., 2006). In this study, Daniel presented his unique and not pre-defined ways of expressing his own play behaviors during play activities, such as sorting and tapping fish in Playset 2 and engaging with treasure toys in Playset 3 (identified within the *individual expression* theme).

Limitations and Future Directions

There are some limitations that could be addressed in future studies. The first limitation pertains to the single participant and the limitations for generality. While SCED methodology quality indicators (Horner et al., 2005; Kratochwill et al., 2013) acknowledge that multiple-baseline/probe design across settings can serve as sufficient evidence of functional relation between the independent and dependent variables, a multiple-baseline/probe design across participants is conventionally used to show the effectiveness of an intervention (e.g., Boudreau & D'Entremont, 2010; Jahr et al., 2000) as a single participant may severely limit generalization. However, we argue that play-based interventions should be individualized to each participant to actually be considered as true play. Future research should explore the extent to which an intervention can be individualized but still be recognized as a single intervention across different participants.

The second limitation is regarding the consideration of playset selections in play-based interventions. In this study, Playset 3 demonstrated a weaker intervention effect compared to the other playsets. Playset 3 emitted a loud rattling sound. The mother later shared that Daniel was sensitive to such noises. Future research should prioritize gathering detailed participant information that could interrupt or reduce play experiences or induce specific interfering behaviors.

The third limitation pertains to our qualitative interaction analysis. While our approach of selecting Playset 2 during maintenance was to capture the maximum skill and play experience of the participant, it may have restricted our comprehension of the participant's holistic growth throughout the intervention. Future work should delve into qualitative analysis across varying conditions (e.g., baseline, generalization conditions) and expand to different play scenarios (e.g., with different playsets in different play environments).

The fourth limitation of the current study pertains to the extent to which the current study adheres to certain quality indicators for SCED. First, our IOA data showed that during some instances, the IOA was below 80% (in 2 out of 16 sessions coded for IOA) albeit with an overall high IOA of 93% for scripted actions and 91% for scripted verbalizations. The authors chose to include all the data and the relatively lower IOA to maintain transparency in the data. Second, we did not assess procedural fidelity. Examining procedure fidelity data could shed light on whether the observed weaker generalization of target behaviors, in comparison to the intervention conditions, might be partly attributed to lower procedure fidelity in the generalization condition, in which Abigail (i.e., Daniel's twin sister) was the play partner rather than the first author. Although these limitations may weaken the methodological strength of a SCED, we argue that these were necessary to allow some degree of freedom in participant behavior to foster natural play and to allow the collection of meaningful qualitative data. For example, some of the low IOA may be a representation of the heterogeneity in how play is defined across individuals, emphasizing the importance of including qualitative analysis in behavioral studies to provide a comprehensive and more nuanced investigation of play. Currently, there are very few methodological recommendations for mixed-methods research especially between SCED and qualitative analysis. In order to establish best practices for mixed-methods research that can evaluate an intervention for use in individuals with autism, more research should be conducted to learn optimal ways to maintain the balance between quantitative and qualitative methods while retaining methodological rigor for both methodologies.

Conclusion

This study supports the evidence for integrating *VM* and *ISM* strategies in enhancing the scripted play behaviors of a preschooler with autism. Moreover, the interaction analysis revealed the participant's unscripted play experiences. These play behaviors are evidence of growth that would have been missed if the effectiveness of the intervention had solely been based on predetermined target behaviors. This exploratory study contributes to play research for individuals with autism through its mixed-method analysis, providing a much richer opportunity to understand how children with autism engage, develop, and find enjoyment in play. We hope this work will encourage future researchers to adopt a broader perspective in exploring children with autism learning and growing through play, leading to interventions that enhance their well-being and quality of life.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10864-024-09550-y>.

Acknowledgements We would like to thank the child with autism and his family who participated in this study. We would also like to thank Xuejing Liu for her research assistance. The authors used person-first language in this paper following the participants' wishes, but we also acknowledge and respect the preference of identity-first language within the autism community.

Declarations

Ethical Approval This study was approved by the Institutional Review Board of the University of Minnesota (STUDY00016518). This study received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of Interest The authors do not have any conflicts of interest to disclose.

References

- Adair, J. K., Phillips, L., Ritchie, J., & Sachdeva, S. (2017). Civic action and play: Examples from Maori, Aboriginal Australian and Latino communities. *Early Child Development and Care*, 187(5–6), 798–811. <https://doi.org/10.1080/03004430.2016.1237049>
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). American Psychiatric Association.
- Ayres, K. M., & Langone, J. (2005). Intervention and instruction with video for students with autism: A review of the literature. *Education and Training in Developmental Disabilities*, 40, 183–196.
- Bak, M. Y. S., Dueñas, A. D., Reilly, A. M., Pan, Q., & Abas, H. (2023). The demographics of autistic individuals in social communication evidence-based practice: A systematic review. *Review Journal of Autism and Developmental Disorders*. <https://doi.org/10.1007/s40489-023-00424-z>
- Bellini, S., Akullian, J., & Hopf, A. (2007). Increasing social engagement in young children with autism spectrum disorders using video self-modeling. *School Psychology Review*, 36(1), 80–90. <https://doi.org/10.1080/02796015.2007.12087953>
- Bernard-Opitz, V., Ing, S., & Kong, T. Y. (2004). Comparison of behavioural and natural play interventions for young children with autism. *Autism*, 8, 319–333. <https://doi.org/10.1177/1362361304045212>
- Bezemer, J., & Mavers, D. (2011). Multimodal transcription as academic practice: A social semiotic perspective. *International Journal of Social Research Methodology*, 14(3), 191–206. <https://doi.org/10.1080/13645579.2011.563616>
- Bottema-Beutel, K., Crowley, S., & Kim, S. Y. (2022). Sequence organization of autistic children's play with caregivers: Rethinking follow-in directives. *Autism*, 26(5), 1011–1311. <https://doi.org/10.1177/13623613211046799>
- Boudreau, E., & D'Entremont, B. (2010). Improving the pretend play skills of preschoolers with autism spectrum disorders: The effects of video modeling. *Journal of Developmental and Physical Disabilities*, 22, 415–431. <https://doi.org/10.1007/s10882-010-9201-5>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- Braun, V., & Clarke, V. (2021). One size fits all? What counts as quality practice in (reflexive) thematic analysis? *Qualitative Research in Psychology*, 18(3), 328–352. <https://doi.org/10.1080/14780887.2020.1769238>
- Burghardt, G. M. (2012). Defining and recognizing play. In A. D. Pellegrini (Ed.), *The Oxford handbook of the development of play*. Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780195393002.013.0002>
- Burney, V., Arnold-Saritepe, A., & McCann, C. M. (2023). Rethinking the place of qualitative methods in behavior analysis. *Perspectives on Behavior Science*, 46, 1–16. <https://doi.org/10.1007/s40614-022-00362-x>

- Campbell, J. L., Quincy, C., Osserman, J., & Pedersen, O. K. (2013). Coding in-depth semistructured interviews: Problems of unitization and intercoder reliability and agreement. *Sociological Methods and Research*, 42(3), 294–320. <https://doi.org/10.1177/0049124113500475>
- Chang, Y.-C., Shire, S., Shih, W., Gelfand, C., & Kasari, C. (2016). Preschool deployment of evidence-based social communication intervention: JASPER in the classroom. *Journal of Autism and Developmental Disorders*, 46(6), 2211–2223. <https://doi.org/10.1007/s10803-016-2752-2>
- Charlop-Christy, M. H., Le, L., & Freeman, K. A. (2000). A comparison of video modeling with in vivo modeling for teaching children with autism. *Journal of Autism and Developmental Disorders*, 30, 537–552.
- Charman, T., Swettenham, J., Baron-Cohen, S., Cox, A., Baird, G., & Drew, A. (1997). Infants with autism: An investigation of empathy, pretend play, joint attention, and imitation. *Developmental Psychology*, 33(5), 781–789. <https://doi.org/10.1037/0012-1649.33.5.781>
- Chen, R. (2022). Improvisations in the embodied interactions of a non-speaking autistic child and his mother: Practices for creating intersubjective understanding. *Cognitive Linguistics*, 33(1), 155–191. <https://doi.org/10.1515/cog-2021-0047>
- D’Ateno, P., Mangiapanello, K., & Taylor, B. A. (2003). Using video modeling to teach complex play sequences to a preschooler with autism. *Journal of Positive Behavior Interventions*, 5(1), 5–11. <https://doi.org/10.1177/10983007030050010801>
- de Leeuw, A., Happé, F., & Hoekstra, R. A. (2020). A conceptual framework for understanding the cultural and contextual factors on autism across the globe. *Autism Research*, 13(7), 1029–1050. <https://doi.org/10.1002/aur.2276>
- De León, L. (2007). Parallelism, metalinguistic play, and the interactive emergence of Zinacantec Mayan siblings’ culture. *Research on Language and Social Interaction*, 40(4), 405–436. <https://doi.org/10.1080/08351810701471401>
- de Ruiter, J. P., & Albert, S. (2017). An appeal for a methodological fusion of conversation analysis and experimental psychology. *Research on Language and Social Interaction*, 50(1), 90–107. <https://doi.org/10.1080/08351813.2017.1262050>
- DeLiema, D., Enyedy, N., & Danish, J. A. (2019). Roles, rules, and keys: How different play configurations shape collaborative science inquiry. *Journal of the Learning Sciences*, 28(4–5), 513–555. <https://doi.org/10.1080/10508406.2019.1675071>
- DeLiema, D., Enyedy, N., Steen, F., & Danish, J. A. (2021). Integrating viewpoint and space: How lamination across gesture, body movement, language, and material resources shapes learning. *Cognition and Instruction*, 39(3), 328–365. <https://doi.org/10.1080/07370008.2021.1928133>
- den Houting, J., Higgins, J., Isaacs, K., Mahony, J., & Pellicano, E. (2021). ‘I’m not just a guinea pig’: Academic and community perceptions of participatory autism research. *Autism*, 25(1), 148–163. <https://doi.org/10.1177/1362361320951696>
- Derry, S. J., Pea, R. D., Barron, B., Engle, R. A., Erickson, F., Goldman, R., Hall, R., Koschmann, T., Lemke, J. L., Sherin, M. G., & Sherin, B. L. (2010). Conducting video research in the learning sciences: Guidance on selection, analysis, technology, and Ethics. *Journal of the Learning Sciences*, 19(1), 3–53. <https://doi.org/10.1080/10508400903452884>
- Dijkstra-de Neijs, L., Tisseur, C., Kluwen, L. A., van Berckelaer-Onnes, I. A., Swaab, H., & Ester, W. A. (2021). Effectivity of play-based interventions in children with autism spectrum disorder and their parents: A systematic review. *Journal of Autism and Developmental Disorders*, 53, 1–30. <https://doi.org/10.1007/s10803-021-05357-2>
- Dindar, K., Korkiakangas, T., Laitila, A., & Kärnä, E. (2017). An interactional “live eye tracking” study in autism spectrum disorder: Combining qualitative and quantitative approaches in the study of gaze. *Qualitative Research in Psychology*, 14(3), 239–265. <https://doi.org/10.1080/14780887.2017.1290174>
- Dueñas, A. D., Plavnick, J. B., & Bak, M. S. (2019). Effects of joint video modeling on unscripted play behavior of children with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 49, 236–247. <https://doi.org/10.1007/s10803-018-3719-2>
- Freeman, S., & Kasari, C. (2013). Parent–child interactions in autism: Characteristics of play. *Autism*, 17(2), 147–161. <https://doi.org/10.1177/1362361312469269>
- Garvey, C. (1974). Some properties of social play. *Merrill-Palmer Quarterly of Behavior and Development*, 20(3), 163–180.
- Gibson, J. L., Pritchard, E., & de Lemos, C. (2021). Play-based interventions to support social and communication development in autistic children aged 2–8 years: A scoping review. *Autism and Developmental Language Impairments*. <https://doi.org/10.1177/23969415211015840>

- Glaser, B. G. (1965). The constant comparative method of qualitative analysis. *Social Problems*, 12(4), 436–445. <https://doi.org/10.2307/798843>
- Gordon, G. (2014). Well played: The origins and future of playfulness. *American Journal of Play*, 6(2), 234–266.
- Graetz, J. E., Matropieri, M. A., & Scruggs, T. E. (2006). Show time: Using video self-modeling to decrease inappropriate behavior. *Teaching Exceptional Children*, 38, 43–48.
- Gray, P. (2019). Evolutionary functions of play: Practice, resilience, innovation, and cooperation. In P. Smith & J. Roopnarine (Eds.), *The Cambridge handbook of play: Developmental and disciplinary perspectives* (pp. 84–102). Cambridge University Press.
- Hall, R., & Stevens, R. (2015). Interaction analysis approaches to knowledge in use. In A. A. diSessa, M. Levin, & N. J. S. Brown (Eds.), *Knowledge and interaction* (pp. 88–124). Routledge.
- Henning, B., Cordier, R., Wilkes-Gillan, S., & Falkmer, T. (2016). A pilot play-based intervention to improve the social play interactions of children with autism spectrum disorder and their typically developing playmates. *Australian Occupational Therapy Journal*, 63(4), 223–232. <https://doi.org/10.1111/1440-1630.12285>
- Hoey, E. M. (2018). How speakers continue with talk after a lapse in conversation. *Research on Language and Social Interaction*, 51(3), 329–346. <https://doi.org/10.1080/08351813.2018.1485234>
- Hoey, E. M., & Kendrick, K. H. (2017). Conversation analysis. In A. M. B. de Groot & P. Hagoort (Eds.), *Research methods in psycholinguistics: A practical guide* (pp. 151–173). Wiley Blackwell.
- Hong, E. R., Ganz, J. B., Morin, K., Davis, J. L., Ninci, J., Neely, L. C., Boles, M. B., & Gilliland, W. D. (2016). The effects of video modeling in teaching functional living skills to person with ASD: A meta-analysis of single-case studies. *Research in Developmental Disabilities*, 57, 158–169. <https://doi.org/10.1016/j.ridd.2016.07.001>
- Horner, R. H., Carr, E. G., Halle, J., McGee, G., Odom, S., & Wolery, M. (2005). The use of single-subject research to identify evidence-based practice in special education. *Exceptional Children*, 71(2), 165–179. <https://doi.org/10.1177/00144029050710020>
- Howard, J., & McInnes, K. (2013). *The essence of play: A practice companion for professionals working with children and young people*. Routledge.
- Hughes, F. P. (2021). *Children, play, and development*. SAGE Publications.
- Hume, K., Steinbrenner, J. R., Odom, S. L., Morin, K. L., Nowell, S. W., Tomaszewski, B., Szendrey, S., McIntyre, N. S., Yücesoy-Özkan, S., & Savage, M. N. (2021). Evidence-based practices for children, youth, and young adults with autism: Third generation review. *Journal of Autism and Developmental Disorders*. <https://doi.org/10.1007/s10803-020-04844-2>
- Jahr, E., Eldevik, S., & Eikeseth, S. (2000). Teaching children with autism to initiate and sustain cooperative play. *Research in Developmental Disabilities*, 21(2), 151–169. [https://doi.org/10.1016/S0891-4222\(00\)00031-7](https://doi.org/10.1016/S0891-4222(00)00031-7)
- Jarrold, C. (2003). A review of research into pretend play in autism. *Autism*, 7(4), 379–390. <https://doi.org/10.1177/1362361303007004004>
- Jordan, B., & Henderson, A. (1995). Interaction analysis: Foundations and practice. *The Journal of the Learning Sciences*, 4(1), 39–103. https://doi.org/10.1207/s15327809jls0401_2
- Jordan, R. (2003). Social play and autistic spectrum disorders: A perspective on theory, implications and educational approaches. *Autism*, 7(4), 347–360. <https://doi.org/10.1177/1362361303007004002>
- Kamps, D., Thiemann-Bourque, K., Heitzman-Powell, L., Schwartz, I., Rosenberg, N., Mason, R., & Cox, S. (2015). A comprehensive peer network intervention to improve social communication of children with autism spectrum disorders: A randomized trial in kindergarten and first grade. *Journal of Autism and Developmental Disorders*, 45, 1809–1824. <https://doi.org/10.1007/s10803-014-2340-2>
- Kasari, C., Chang, Y. C., & Patterson, S. (2013). Pretending to play or playing to pretend: The case of autism. *American Journal of Play*, 6(1), 124.
- Kent, C., Cordier, R., Joosten, A., Wilkes-Gillan, S., Bundy, A., & Speyer, R. (2020). A systematic review and meta-analysis of interventions to improve play skills in children with autism spectrum disorder. *Review Journal of Autism and Developmental Disorders*, 7(1), 91–118. <https://doi.org/10.1007/s40489-019-00181-y>
- Kim, S., & Clarke, E. (2015). Case study: An iPad-based intervention on turn-taking behaviors in preschoolers with autism. *Behavioral Development Bulletin*, 20(2), 253. <https://doi.org/10.1037/h0101314>
- Knight, V. F., Spooner, F., Browder, D. M., Smith, B. R., & Wood, C. L. (2013). Using systematic instruction and graphic organizers to teach science concepts to students with autism spectrum disorders

- and intellectual disability. *Focus on Autism and Other Developmental Disabilities*, 28(2), 115–126. <https://doi.org/10.1177/1088357612475301>
- Korkiakangas, T., & Rae, J. (2014). The interactional use of eye-gaze in children with autism spectrum disorders. *Interaction Studies*, 15(2), 233. <https://doi.org/10.1075/is.15.2.12kor>
- Kratochwill, T. R., Hitchcock, J. H., Horner, R. H., Levin, J. R., Odom, S. L., Rindskopf, D. M., & Shadish, W. R. (2013). Single-case intervention research design standards. *Remedial and Special Education*, 34(1), 26–38. <https://doi.org/10.1177/07419325124527>
- Lane, J. D., & Gast, D. L. (2014). Visual analysis in single case experimental design studies: Brief review and guidelines. *Neuropsychological Rehabilitation*, 24(3–4), 445–463. <https://doi.org/10.1080/09602011.2013.815636>
- Lang, R., O'Reilly, M., Rispoli, M., Shogren, K., Machalicek, W., Sigafoos, J., & Regeher, A. (2009). Review of interventions to increase functional and symbolic play in children with autism. *Education and Training in Developmental Disabilities*, 44, 481–492.
- Ledford, J. R., Lambert, J. M., Pustejovsky, J. E., Zimmerman, K. N., Hollins, N., & Barton, E. E. (2023). Single-case-design research in special education: Next-generation guidelines and considerations. *Exceptional Children*, 89(4), 379–396. <https://doi.org/10.1177/0014402922113765>
- Lee, J. N. (2015). The effectiveness of point-of-view video modeling as a social skills intervention for children with autism spectrum disorders. *Review Journal of Autism and Developmental Disorders*, 2, 414–428. <https://doi.org/10.1007/s40489-015-0061-x>
- Levin, M., & Brown, N. J. (Eds.). (2015). *Knowledge and interaction: A synthetic agenda for the learning sciences*. Routledge.
- Lord, C., Rutter, M., DiLavore, P. C., Risi, S., Gotham, K., & Bishop, S. (2012). *Autism diagnostic observation schedule, second edition (ADOS-2)*. Western Psychological Services.
- Luckey, A. J., & Fabes, R. A. (2005). Understanding nonsocial play in early childhood. *Early Childhood Education Journal*, 33, 67–72. <https://doi.org/10.1007/s10643-006-0054-6>
- Ma, X. E., Gu, H., & Zhao, J. (2021). Atypical gaze patterns to facial feature areas in autism spectrum disorders reveal age and culture effects: A meta-analysis of eye-tracking studies. *Autism Research*, 14(12), 2625–2639. <https://doi.org/10.1002/aur.2607>
- MacDonald, R., Clark, M., Garrigan, E., & Vangala, M. (2005). Using video modeling to teach pretend play to children with autism. *Behavioral Interventions*, 20, 225–238. <https://doi.org/10.1002/bin.197>
- Maione, L., & Mirenda, P. (2006). Effects of video modeling and video feedback on peer-directed social language skills of a child with autism. *Journal of Positive Behavior Interventions*, 8, 106–118. <https://doi.org/10.1177/10983007060080020201>
- McCoy, A., Holloway, J., Healy, O., Rispoli, M., & Neely, L. (2016). A systematic review and evaluation of video modeling, role-play and computer-based instruction as social skills interventions for children and adolescents with high-functioning autism. *Review Journal of Autism and Developmental Disorders*, 3, 48–67. <https://doi.org/10.1007/s40489-015-0065-6>
- McCoy, K., & Hermansen, E. (2007). Video modeling for individuals with autism: A review of model types and effects. *Education and Treatment of Children*, 30, 183–213. <https://doi.org/10.1353/etc.2007.0029>
- McCrudden, M. T., Marchand, G., & Schutz, P. (2019). Mixed methods in educational psychology inquiry. *Contemporary Educational Psychology*, 57, 1–8. <https://doi.org/10.1016/j.cedpsych.2019.01.008>
- Nikopoulos, C. K., & Keenan, M. (2003). Promoting social initiation in children with autism using video modeling. *Behavioral Interventions: Theory and Practice in Residential and Community-Based Clinical Programs*, 18(2), 87–108. <https://doi.org/10.1002/bin.129>
- O'Brien, J., Coker, P., Lynn, R., Suppinger, R., Pearigen, T., Rabon, S., St. Aubin, M., & Ward, A. T. (2000). The impact of occupational therapy on a child's playfulness. *Occupational Therapy in Health Care*, 12(2–3), 39–51. https://doi.org/10.1080/J003v12n02_0
- O'Keeffe, C., & McNally, S. (2023). A systematic review of play-based interventions targeting the social communication skills of children with autism spectrum disorder in educational contexts. *Review Journal of Autism and Developmental Disorders*, 10(1), 51–81. <https://doi.org/10.1007/s40489-021-00286-3>
- Pane, H. M., Sidener, T. M., Reeve, S. A., Kisamore, A., & Nirgudkar, A. (2022). A comparison of development-matched and age-matched targets on play skills of children with autism spectrum disorder. *Journal of Applied Behavior Analysis*, 55(1), 195–213. <https://doi.org/10.1002/jaba.891>
- Paterson, C. R., & Arco, L. (2007). Using video modeling for generalizing toy play in children with autism. *Behavior Modification*, 31(5), 660–681. <https://doi.org/10.1177/0145445507301651>

- Perepa, P. (2014). Cultural basis of social ‘deficits’ in autism spectrum disorders. *European Journal of Special Needs Education*, 29(3), 313–326. <https://doi.org/10.1080/08856257.2014.908024>
- Qi, C. H., Barton, E. E., Collier, M., & Lin, Y.-L. (2018). A systematic review of single-case research studies on using video modeling interventions to improve social communication skills for individuals with autism spectrum disorder. *Focus on Autism and Other Developmental Disabilities*, 33(4), 249–257. <https://doi.org/10.1177/1088357617741282>
- Rieth, S. R., Stahmer, A. C., Suhrheinrich, J., Schreibman, L., Kennedy, J., & Ross, B. (2014). Identifying critical elements of treatment: Examining the use of turn taking in autism intervention. *Focus on Autism and Other Developmental Disabilities*, 29(3), 168–179.
- Ruberg, B. (2017). *Playing to lose: The queer art of failing at video games*. Indiana University Press.
- Rutherford, M. D., Young, G. S., Hepburn, S., & Rogers, S. J. (2007). A longitudinal study of pretend play in autism. *Journal of Autism and Developmental Disorders*, 37, 1024–1039. <https://doi.org/10.1007/s10803-006-0240-9>
- Sacks, H., Schegloff, E. A., & Jefferson, G. (1978). *A simplest systematics for the organization of turn taking for conversation*. Academic Press.
- Schegloff, E. A. (1993). Reflections on quantification in the study of conversation. *Research on Language and Social Interaction*, 26(1), 99–128. https://doi.org/10.1207/s15327973rlsi2601_5
- Sicart, M. (2014). *Play matters*. MIT Press.
- Sidnell, J. (2011). The epistemics of make-believe. In T. Stivers, L. Mondada, & J. Steensig (Eds.), *The morality of knowledge in conversation* (pp. 131–155). Cambridge University Press.
- Siraj-Blatchford, I. (2009). Conceptualising progression in the pedagogy of play and sustained shared thinking in early childhood education: A Vygotskian perspective. *Educational and Child Psychology*, 26, 77–89. <https://doi.org/10.53841/bpsecp.2009.26.2.77>
- Skaines, N., Rodger, S., & Bundy, A. (2006). Playfulness in children with autistic disorder and their typically developing peers. *British Journal of Occupational Therapy*, 69(11), 505–512. <https://doi.org/10.1177/030802260606901104>
- Smith, T. (2001). Discrete trial training in the treatment of autism. *Focus on Autism and Other Developmental Disabilities*, 16(2), 86–92. <https://doi.org/10.1177/108835760101600204>
- Souto-Manning, M. (2017). Is play a privilege or a right? And what’s our responsibility? On the role of play for equity in early childhood education. *Early Child Development and Care*, 187(5–6), 785–787. <https://doi.org/10.1080/03004430.2016.1266588>
- Trevisan, D. A., Roberts, N., Lin, C., & Birmingham, E. (2017). How do adults and teens with self-declared autism spectrum disorder experience eye contact? A qualitative analysis of first-hand accounts. *PLoS ONE*, 12(11), e0188446. <https://doi.org/10.1371/journal.pone.0188446>
- Therrien, M. C., & Light, J. C. (2018). Promoting peer interaction for preschool children with complex communication needs and autism spectrum disorder. *American Journal of Speech-Language Pathology*, 27(1), 207–221. https://doi.org/10.1044/2017_AJSLP-17-0104
- Wiemann, J. M., & Knapp, M. L. (2017). Turn-taking in conversations. *Communication Theory*, 4, 226–245. <https://doi.org/10.4324/9781315080918-19>
- Wright, J. C., Knight, V. F., & Barton, E. E. (2020). A review of video modeling to teach STEM to students with autism and intellectual disability. *Research in Autism Spectrum Disorder*, 70, 101476. <https://doi.org/10.1016/j.rasd.2019.101476>

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